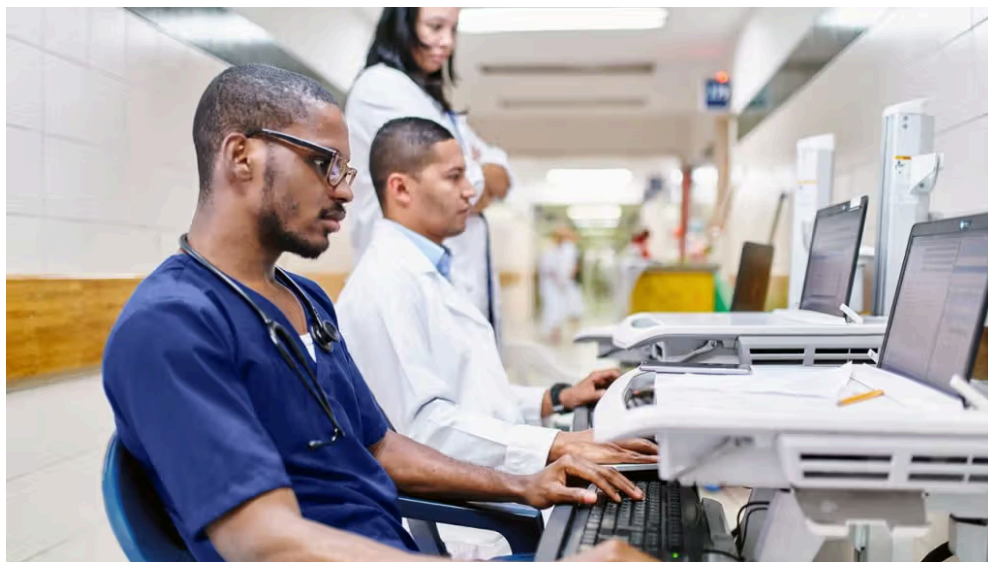


Melanoma: AI shows promise, but real-world use raises questions



Evidence suggests that AI tools assisting dermatologists can improve diagnostic performance for skin cancer. Image credit: PER Images/Stocksy

- A study suggests that artificial intelligence (AI) showed comparable performance to dermatologists in detecting melanoma, with similar sensitivity and specificity in prospective clinical studies.
- However, AI demonstrated a higher specificity in direct comparisons, suggesting it may help reduce unnecessary biopsies by better identifying benign lesions.
- The combination of dermatologists using dermoscopy with AI support showed the highest diagnostic performance.
- Despite the promising results, current evidence remains limited and potentially biased, highlighting the need for larger, real-world studies before widely adopting AI in clinical practice.

Experts often consider [melanoma](#) to be the [most serious](#) type of [skin cancer](#) due to the likelihood of [metastasizing](#). Evidence suggests that roughly [112,000 Americans](#)

receive a diagnosis of melanoma each year, and more than [1.5 million people](#)

in the U.S. are living with melanoma.

While only accounting for about 1% of skin cancers, melanoma causes a large [majority](#) of skin cancer deaths.

Early detection of melanoma can be critical, as the 5-year survival rate for early melanoma is [94%](#). A common approach for detection is via a dermatologist's diagnosis. However, reliance on specialists can make it difficult for people to receive a timely diagnosis.

Tools such as a [dermatoscope](#) can greatly [improve](#) the accuracy of skin cancer detection among dermatology clinicians. However, could AI further improve detection rates?

AI performs on par with dermatologists

In recent years, there has been growing interest in AI-based analysis as a [diagnostic aid](#) for melanoma. However, the use of AI in real-world clinical settings remains controversial. While studies suggest that AI could be utilized in these settings, real-world evidence has been limited.

To address this, researchers conducted a systematic review and meta-analysis of 11 prospective studies. These studies involved more than 2,500 participants and 50 dermatologists.

The findings suggest that for melanoma diagnosis, dermatologists achieved a sensitivity of 78.6% and specificity of 75.2%. AI systems alone reached 80.9% sensitivity and 75.6% specificity.

These results suggest that AI demonstrated a diagnostic performance comparable to that of dermatologists.

Notably, in one study, dermatologists assisted by AI achieved 91.9% sensitivity and 83.7% specificity. This indicates that AI could serve as a valuable tool to assist, rather than replace, clinicians.

[Tanya Evans](#), MD, board certified dermatologist and medical director of the Skin Cancer Program at the Melanoma Clinic at MemorialCare Saddleback Medical Center in Laguna Hills, CA, who was not involved in the study, notes the key takeaway is not that AI could replace dermatologists, but can reliably function as a clinical adjunct:

“The most important implication is that diagnostic accuracy can be meaningfully improved when AI is combined with clinician expertise, rather than used independently. The study emphasizes that AI is still early in validation, with bias and limited generalizability—so it is not yet ready for autonomous use.”

— Tanya Evans, MD

AI may reduce unnecessary biopsies

While comparable, the AI systems tended to show a higher specificity in head-to-head comparisons within the same clinical settings. This suggests the AI tools were better at correctly identifying benign lesions.

The higher specificity may have practical implications. When uncertain, dermatologists typically apply caution and will recommend biopsies. By contrast, AI may help to reduce unnecessary procedures by ruling out noncancerous lesions.

“This is one of the most promising and immediately actionable implications,” Evans said.

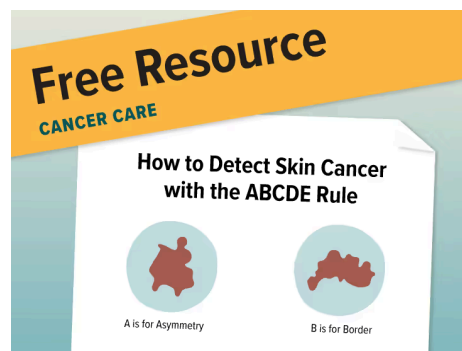
“The most immediate benefit is likely reducing unnecessary biopsies while maintaining safety. Dermatologists tend to be risk-averse (biopsy when uncertain), whereas AI is more probabilistic and specific—this complementary dynamic is where synergy happens. The most realistic near-term model is AI layered onto dermoscopy—not replacing it.”

— Tanya Evans, MD

The research team also suggest that although evidence remains limited, these findings indicate combining human expertise with AI

support could offer the best results.

While the findings support the potential of AI as a decision-support tool, the researchers emphasize that the technology is still in an early validation phase. They add that larger, multicenter studies are still necessary to determine the safety, reliability, and real-world clinical impact of these tools.



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Important limitations

As noted, this review used prospective studies to evaluate performance in real-time clinical settings. This means they should be more reflective of everyday practice, rather than retrospective research, which relies on preselected datasets.

This suggests that AI's performance should remain strong even outside controlled laboratory conditions. However, despite these promising findings, the study is not without its limitations.

The researchers highlight a high risk of bias, as most of the studies reviewed only included lesions already suspected of melanoma, rather than the full range a clinician will see in routine practice.

Additionally, many of the studies used simplified diagnoses, which does not reflect real clinical decision making, and differences in the

study designs and datasets make comparisons challenging.

As such, these factors mean that the results may not fully translate to everyday healthcare settings.

The researchers conclude that although AI shows promise for matching dermatologist performance, more robust evidence is necessary before it becomes a standard tool in clinics. For now though, it may serve as a potential tool for clinicians to help improve early cancer detection.

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